

JESSE DONWAHOUE

Master's Application — Robotics

Embedded Systems & IoT

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LANGUAGES

- **French : native**
- English : technical

(datasheets, docs, tutorials)

SOFT SKILLS

- Fast learner
- Analytical mindset
- Rigor & methodology
- Technical curiosity
- Adaptability
- Works well under pressure

INTERESTS

- Robotics & IoT
- Autonomous systems
- Bioinformatics
- Sport | Music
- Travel

CURRENTLY LEARNING

Protocols

I2C, SPI, UART

Control

PWM, PID (basics)

Actuators

Stepper motors, Dynamixel

Tools

PlatformIO, ROS, Raspberry Pi

PROFILE

Front-end developer with a Software Engineering degree, currently pursuing dual training in embedded electronics and bioinformatics.

I design autonomous robotic systems: sensors → embedded processing (ESP32/Arduino) → actuators.

Fast learner, technical rigor, proven ability to take projects from concept to completion.

EDUCATION

Degree / Training	Institution	Period
Electronics & Embedded IoT Training (ongoing)	ADESTI • Cotonou	Feb. 2026 – present
Bioinformatics Training (ongoing)	CEDORIASE ACADEMY	Apr. 2026 – present
Bachelor's Degree in Software Engineering	IFRI / UAC • Calavi	2021 – 2025
High School Diploma (Baccalaureat)	Complexe Scolaire Planete Educative	2020 – 2021

ROBOTICS & EMBEDDED PROJECTS

Autonomous Scarecrow Robot — 2 Evolutionary Versions ²⁰²⁶

- ▶ Board: Arduino / ESP32
- ▶ Sensor: PIR (motion detection up to 7 m)
- ▶ Actuators:
 - Version 1 (2026): electronic relay-based control
 - Version 2 (2026): electromechanical contactor control (more robust, higher loads)
- ▶ Embedded logic: hardware interrupts + timers, automatic triggering of light / siren / servomotor
- ▶ Power supply: 12V battery, sleep-mode power consumption optimization
- ▶ Language: C++ (Arduino IDE)

This project demonstrates my ability to design an autonomous, scalable and robust system under real-world conditions.

Real-Time Electrical Consumption Monitoring System (IoT) ²⁰²⁶

- ▶ Board: ESP32 (built-in Wi-Fi)
- ▶ Sensors: voltage and current (real-time measurement)
- ▶ Pipeline: sensors → ESP32 → embedded web interface (HTML/CSS/JS)
- ▶ Automatic email alerts via SMTP on overconsumption detection

- Board: Arduino
- Sensors: HC-SR04 ultrasonic (per-slot presence detection)
- Outputs: LED display + signaling panel
- Logic: automated real-time slot availability management

TECHNICAL SKILLS — ROBOTICS & EMBEDDED

Microcontrollers	ESP32, Arduino (C/C++)
Sensors	PIR, ultrasonic HC-SR04, current/voltage, temperature, LDR
Actuators	Relays, contactors, LEDs, buzzer, servomotor (basics)
Embedded	Interrupts, timers, sleep management, C++
Communication	Wi-Fi (ESP32), I2C, SPI, UART (learning)
Web Dev	HTML/CSS/JS, Vue.js, Node.js, REST API, SQL
Tools	Git, PlatformIO, VS Code, Figma

PROFESSIONAL EXPERIENCE

Position	Company	Period
Front-End Developer	ACT • Abomey-Calavi	Mar. – Jul. 2024
Web Developer (intern)	Gits • Cotonou	Aug. – Sep. 2023
Customer Service Agent	Carrefour / Concentrix • Cotonou	Jul. 2025 – Jan. 2026

- ACT: UI design for waste collection tracking app, IoT sensor integration (light, temperature), database modeling
- Gits: REST API development with Node.js/Express.js, dynamic web interfaces
- Concentrix: Call/email/message handling, effective communication under pressure